Validating SysJava's seccomp profiles under real workloads, and binding vs no binding

Ten JDK 8 containers (Eclipse Temurin 8u504, Debian 10). Every container ran the same workload three ways (Docker's default seccomp profile, our profile in log-only mode, our profile enforced), three times each: 90 runs. Part 2 compares the static analysis with and without binding recovery for the same ten images.

1. Result overview

Application	Version	Profile (syscalls)	Ops per run	Failed ops (9 runs)	Syscalls outside the profile (log mode, all 3 reps)	Result
Apache ZooKeeper	3.4.14	190	3,951	0	none	PASS
Apache Tomcat	7.0.109	170	3,052	0	fchdir	Works under enforce; profile misses fchdir
Apache Cassandra	3.0.29	230	11	0	none	PASS
Eclipse Jetty	9.4.51.v20230217	157	3,052	0	fchdir, fchown	Works under enforce; profile misses fchdir, fchown
Apache Storm	2.4.0	172	35	0	fallocate, fchdir	Works under enforce; profile misses fallocate, fchdir
OrientDB	3.2.19	173	1,808	0	none	PASS
Elasticsearch	7.5.2	183	250	0	fchown	Works under enforce; profile misses fchown
Apache Solr	8.7.0	199	199	0	times	Works under enforce; profile misses times
Apache Groovy	4.0.0	189	46	0	none	PASS
JRuby	9.4.2.0	167	46	0	none	PASS

These are the profiles the paper reports. Section 6 repeats all 90 runs with the profiles after three analysis fixes for the syscalls found here: all ten applications then pass.

2. How the validation works

Images. The same ten images SysJava analysed (the paper's JDK 8 table). Each profile is the one the paper reports: `Final_results_JDK8/<image>/Binary_Analysis/syscalls_output_<image>/<image>.json`; for Storm the 172-syscall profile without the *sync* that came only from false FFM hits.

Application	Image	Version reported by the container	JDK	OS
Apache ZooKeeper	zookeeper:3.4.14-jdk8-buster	3.4.14	Temurin 1.8.0_504	Debian 10 (buster)
Apache Tomcat	tomcat:7.0-jdk8-buster	7.0.109	Temurin 1.8.0_504	Debian 10 (buster)
Apache Cassandra	cassandra:3.0.29-jdk8-buster	3.0.29	Temurin 1.8.0_504	Debian 10 (buster)
Eclipse Jetty	jetty:9.4.51-jdk8-buster	9.4.51.v20230217	Temurin 1.8.0_504	Debian 10 (buster)
Apache Storm	storm:2.4.0-jdk8-buster	2.4.0	Temurin 1.8.0_504	Debian 10 (buster)
OrientDB	orientdb:3.2.19-jdk8-buster	3.2.19	Temurin 1.8.0_504	Debian 10 (buster)
Elasticsearch	elasticsearch:7.5.2-jdk8-buster	7.5.2	Temurin 1.8.0_504	Debian 10 (buster)
Apache Solr	solr:8.7.0-jdk8-buster	8.7.0	Temurin 1.8.0_504	Debian 10 (buster)

Application	Image	Version reported by the container	JDK	OS
Apache Groovy	groovy:4.0.0-jdk8-buster	4.0.0	Temurin 1.8.0_504	Debian 10 (buster)
JRuby	jruby:9.4.2.0-jdk8-buster	9.4.2.0	Temurin 1.8.0_504	Debian 10 (buster)

Three modes, same workload.

Mode	Seccomp profile	What it tells us
base	Docker's default profile (no --security-opt)	The workload itself works: the reference number of operations.
log	Our profile with defaultAction changed from SCMP_ACT_ERRNO to SCMP_ACT_LOG	Nothing is blocked. Every syscall the profile would deny is allowed but written to the audit log (type=SECCOMP, code=0x7ffc0000). This shows denials that an application might tolerate silently (an EPERM it ignores).
enforce	Our profile exactly as published (--security-opt seccomp=<image>.json)	What a user gets: anything outside the profile fails with EPERM.

Order and repetitions. For each repetition 1 to 3 the harness runs base, then log, then enforce, each in a fresh container (name `wlv-<app>-<mode>-<rep>`), and removes it afterwards. Different applications ran in parallel (120 GB host).

Who runs the workload. Only the application runs inside the confined container. Requests come from the host (HTTP/REST, TCP) or from a second, unconfined container of the same image (cqlsh and cassandra-stress for Cassandra, zkCli for ZooKeeper, the storm client for Storm). So the profile is tested on the application's own syscalls, not on the test tools'. Groovy and JRuby are language runtimes with no server; there the workload is a script the container runs as its program.

Counting operations. Every request or command is one operation and is checked: HTTP status 200 (or 404 for an undeployed app), command exit code 0, cassandra-stress reaching END with no error, zkCli output without exceptions. The batch scripts check each step themselves (e.g. 200 files written and listed back) and print `OPS ok=N fail=N`.

Detecting would-be denials in log mode. A thread polls `docker top <container> -eo pid` every second and collects the container's host PIDs. After the run, the harness reads every SCMP_ACT_LOG record (type=SECCOMP, code=0x7ffc0000) written since the container started from the audit log, and keeps the ones whose pid (the thread group of the process) belongs to the container. Records of other processes are kept apart as unattributed. The syscall number is translated with `ausyscall x86_64`.

Positive control. To show the detection works, JRuby ran a three-line script (mkdir, write, rename) under its log profile with *mkdir* and *rename* removed. The program completed ("done") and the harness reported: **mkdir (java), rename (java)**; unattributed: none.

PASS means, in all three repetitions: the application became ready in every mode, enforce completed exactly as many operations as base, no operation failed, and log mode recorded no syscall outside the profile. When everything works under enforce but log mode records syscalls outside the profile, the result says "Works under enforce; profile misses ...".

3. Per-application workloads and results

Apache ZooKeeper 3.4.14 *(zookeeper:3.4.14-jdk8-buster, profile 190 syscalls)*

Server in the confined container. A client container (same image, unconfined) runs zkCli; the host sends the four-letter admin commands over TCP. **Ready when:** port 2181 open and "binding to port" in the log.

Workload step	Ops per run	Exercises
Four-letter commands ruok, stat, srvr, conf, cons, envi, mntr, wchs, dump (x3)	27	admin socket, server stats, JVM environment
zkCli round: create /wl, 400 creates, 400 sets, 100 gets, ls, stat, sync, sequential and ephemeral nodes, 400 deletes, rmr	3 x 1,308	transaction log writes, snapshots, sessions, watches

Rep	Mode	Ready after (s)	Workload (s)	Ops ok	Ops failed	Outside the profile (log mode)
1	base	6	8	3,951	0	n/a
1	log	7	8	3,951	0	none
1	enforce	6	7	3,951	0	(not measured)
2	base	6	7	3,951	0	n/a
2	log	6	9	3,951	0	none
2	enforce	6	7	3,951	0	(not measured)
3	base	6	7	3,951	0	n/a
3	log	6	9	3,951	0	none
3	enforce	6	6	3,951	0	(not measured)

Result: PASS

Apache Tomcat 7.0.109 *(tomcat:7.0-jdk8-buster, profile 170 syscalls)*

Server in the confined container with the stock webapps (examples, docs, ROOT; no manager apps) mounted as its webapps folder. All requests come from the host. **Ready when:** GET /examples/ returns 200.

Workload step	Ops per run	Exercises
GET on 12 servlet, JSP and static pages (HelloWorld, RequestInfo, RequestHeader, Cookie, EL, simple tag, date, numguess, session cart, docs) x250, 16 in parallel	3000	servlet dispatch, JSP compilation (javac in the JVM), sessions, static files
POST form to RequestParamExample	50	request body parsing
Hot deploy: copy wl.war into webapps, wait for /wl/ = 200	1	auto-deployer: unpack war, new class loader
Undeploy: delete wl.war, wait for /wl/ = 404	1	directory delete (rmdir/unlink)

Rep	Mode	Ready after (s)	Workload (s)	Ops ok	Ops failed	Outside the profile (log mode)
1	base	3	21	3,052	0	n/a
1	log	4	19	3,052	0	fchdir (java)
1	enforce	3	19	3,052	0	(not measured)
2	base	3	19	3,052	0	n/a
2	log	4	20	3,052	0	fchdir (java)
2	enforce	3	21	3,052	0	(not measured)
3	base	3	21	3,052	0	n/a
3	log	3	21	3,052	0	fchdir (java)
3	enforce	3	19	3,052	0	(not measured)

Result: Works under enforce; profile misses fchdir

Apache Cassandra 3.0.29 *(cassandra:3.0.29-jdk8-buster, profile 230 syscalls)*

Database in the confined container (1 GB heap). cqlsh and cassandra-stress run in a second, unconfined container of the same image. Each line below counts as one operation. **Ready when:** "Starting listening for CQL clients" in the log, then 10 s.

Workload step	Ops per run	Exercises
cqlsh: CREATE KEYSPACE, CREATE TABLE, CREATE INDEX, BATCH of 200 inserts, count, indexed SELECT, TRUNCATE, DROP KEYSPACE	8	schema changes, commit log, memtables, secondary index, truncate snapshot
cassandra-stress write n=60,000, 16 threads	1	commit log, memtable flush, SSTable writes, compaction
cassandra-stress read n=60,000	1	SSTable reads, caches
cassandra-stress mixed write:read 1:3 n=60,000	1	concurrent reads and writes

Rep	Mode	Ready after (s)	Workload (s)	Ops ok	Ops failed	Outside the profile (log mode)
1	base	13	50	11	0	n/a
1	log	16	55	11	0	none
1	enforce	16	50	11	0	(not measured)
2	base	12	52	11	0	n/a
2	log	12	48	11	0	none
2	enforce	13	47	11	0	(not measured)
3	base	16	54	11	0	n/a
3	log	12	48	11	0	none
3	enforce	12	47	11	0	(not measured)

Result: PASS

Eclipse Jetty 9.4.51.v20230217 *(jetty:9.4.51-jdk8-buster, profile 157 syscalls)*

Server in the confined container with a test war (Tomcat's ROOT app) mounted in its webapps folder. This Jetty image has no JSP support, so the requests use the war's static files. **Ready when:** GET /wl/tomcat.css returns 200.

Workload step	Ops per run	Exercises
GET on 6 static files of the war x500, 16 in parallel	3000	web app class loading, static content, NIO
POST	50	request handling
Hot deploy of wl2.war, wait for /wl2/ = 200	1	deployment scanner, war unpack
Undeploy (delete the war), wait for 404	1	directory delete

Rep	Mode	Ready after (s)	Workload (s)	Ops ok	Ops failed	Outside the profile (log mode)
1	base	3	6	3,052	0	n/a
1	log	4	8	3,052	0	fchdir (java), fchown (runc)
1	enforce	3	7	3,052	0	(not measured)
2	base	3	7	3,052	0	n/a
2	log	3	8	3,052	0	fchdir (java), fchown (runc)
2	enforce	3	6	3,052	0	(not measured)
3	base	4	6	3,052	0	n/a
3	log	4	7	3,052	0	fchdir (java), fchown (runc)
3	enforce	4	6	3,052	0	(not measured)

Result: Works under enforce; profile misses fchdir, fchown

Apache Storm 2.4.0 *(storm:2.4.0-jdk8-buster, profile 172 syscalls)*

Nimbus in the confined container, on a Docker network with the ZooKeeper container. The storm client runs in a separate unconfined container. **Ready when:** nimbus Thrift port 6627 accepts connections, then 10 s.

Workload step	Ops per run	Exercises
storm list	10	Thrift RPC, ZooKeeper state
blobstore create (upload a file)	5	local blob store writes
blobstore list / cat / update / delete	20	blob reads, replacement, delete

Rep	Mode	Ready after (s)	Workload (s)	Ops ok	Ops failed	Outside the profile (log mode)
1	base	16	90	35	0	n/a
1	log	16	88	35	0	fallocate (java), fchdir (java)
1	enforce	16	90	35	0	(not measured)
2	base	16	88	35	0	n/a
2	log	16	87	35	0	fallocate (java), fchdir (java)
2	enforce	16	91	35	0	(not measured)
3	base	16	92	35	0	n/a
3	log	16	87	35	0	fallocate (java), fchdir (java)
3	enforce	16	90	35	0	(not measured)

Result: Works under enforce; profile misses fallocate, fchdir

OrientDB 3.2.19 *(orientdb:3.2.19-jdk8-buster, profile 173 syscalls)*

Database server in the confined container. All requests use its REST API from the host. **Ready when:** GET /listDatabases returns 200.

Workload step	Ops per run	Exercises
Create database wl (plocal)	1	storage files, WAL
Schema: class Person, two properties, index, edge class Knows	5	schema, index files
INSERT Person	1500	record writes, index updates
CREATE EDGE	200	graph writes
Queries: count, range, point lookup, traversal (x25)	100	reads
UPDATE	1	bulk update
Drop database	1	file and directory delete

Rep	Mode	Ready after (s)	Workload (s)	Ops ok	Ops failed	Outside the profile (log mode)
1	base	7	8	1,808	0	n/a
1	log	7	9	1,808	0	none
1	enforce	3	8	1,808	0	(not measured)
2	base	4	7	1,808	0	n/a
2	log	4	9	1,808	0	none
2	enforce	3	7	1,808	0	(not measured)
3	base	3	8	1,808	0	n/a
3	log	3	10	1,808	0	none
3	enforce	3	7	1,808	0	(not measured)

Result: PASS

Elasticsearch 7.5.2 *(elasticsearch:7.5.2-jdk8-buster, profile 183 syscalls)*

Node in the confined container (1 GB heap). All requests use the REST API from the host. **Ready when:** GET /_cluster/health returns 200.

Workload step	Ops per run	Exercises
Create index (2 shards), twice	2	index directories
_bulk, 1,000 documents each, 20 per index	40	translog, Lucene segments, refresh
Searches: match, range, terms aggregation, sorted term (x25 per index)	200	Lucene reads, mmap
_forcemerge to 1 segment, per index	2	segment merge, file delete
_update_by_query with a script, per index	2	Painless scripting, rewrites
_reindex wl1 -> wl3	1	bulk copy
Delete 3 indices	3	directory delete

Rep	Mode	Ready after (s)	Workload (s)	Ops ok	Ops failed	Outside the profile (log mode)
1	base	16	5	250	0	n/a
1	log	13	8	250	0	fchown (runc)
1	enforce	12	7	250	0	(not measured)
2	base	9	8	250	0	n/a
2	log	13	9	250	0	fchown (runc)
2	enforce	12	12	250	0	(not measured)
3	base	12	7	250	0	n/a
3	log	13	7	250	0	fchown (runc)
3	enforce	9	8	250	0	(not measured)

Result: Works under enforce; profile misses fchown

Apache Solr 8.7.0 ***(solr:8.7.0-jdk8-buster, profile 199 syscalls)***

Server in the confined container. All requests use the HTTP API from the host. **Ready when:** GET /solr/admin/info/system returns 200.

Workload step	Ops per run	Exercises
Create core wl from the _default config set	1	core directory, config load
Index Solr's 14 example XML documents, 5 rounds	70	update handler, transaction log
JSON bulk of 2,000 documents with commit, 5 rounds	5	Lucene segments, commit
Queries: all, term, facet, range+sort, 500 rows, wildcard (x20)	120	search, faceting
Optimize	1	segment merge
Backup through the replication handler	1	snapshot copy
UNLOAD the core, deleting index, data and instance directories	1	recursive delete (rmdir)

Rep	Mode	Ready after (s)	Workload (s)	Ops ok	Ops failed	Outside the profile (log mode)
1	base	3	10	199	0	n/a
1	log	7	11	199	0	times (java)
1	enforce	7	13	199	0	(not measured)
2	base	4	9	199	0	n/a
2	log	4	10	199	0	times (java)
2	enforce	4	10	199	0	(not measured)
3	base	7	9	199	0	n/a
3	log	7	12	199	0	times (java)
3	enforce	3	10	199	0	(not measured)

Result: Works under enforce; profile misses times

Apache Groovy 4.0.0 ***(groovy:4.0.0-jdk8-buster, profile 189 syscalls)***

Batch program: the container runs groovy /wl/workload.groovy (script mounted read-only). The script counts its own steps and prints OPS ok=N fail=N. **Ready when:** (batch: the program's exit).

Workload step	Ops per run	Exercises
5 rounds of: write 200 files; read them with a regex; rename; JSON 2,000 objects; XML 500 elements; zip 200 files; 64 tasks on 8 threads; local HTTP server + 200 requests; delete the tree	5 x 9	file I/O, rename, regex, JSON/XML, zip, threads, sockets, directory delete
Delete the temp directory	1	recursive delete

Rep	Mode	Ready after (s)	Workload (s)	Ops ok	Ops failed	Outside the profile (log mode)
1	base	50	0	46	0	n/a
1	log	50	0	46	0	none

Rep	Mode	Ready after (s)	Workload (s)	Ops ok	Ops failed	Outside the profile (log mode)
1	enforce	50	0	46	0	(not measured)
2	base	50	0	46	0	n/a
2	log	49	0	46	0	none
2	enforce	48	0	46	0	(not measured)
3	base	48	0	46	0	n/a
3	log	48	0	46	0	none
3	enforce	48	0	46	0	(not measured)

For the batch programs, "ready after" is the whole run (start to exit).

Result: PASS

JRuby 9.4.2.0 (jruby:9.4.2.0-jdk8-buster, profile 167 syscalls)

Batch program: the container runs jruby /wl/workload.rb (script mounted read-only). The script counts its own steps and prints OPS ok=N fail=N. **Ready when:** (batch: the program's exit).

Workload step	Ops per run	Exercises
5 rounds of: write 200 files; read with a regex; rename + chmod; JSON 2,000 objects; zlib; 16 Ruby threads; local TCP echo server + 100 connections; Java ArrayList; rm -rf	5 x 9	file I/O through JNR (native POSIX), chmod, zlib, threads, sockets, Java interop
Delete the temp directory	1	recursive delete

Rep	Mode	Ready after (s)	Workload (s)	Ops ok	Ops failed	Outside the profile (log mode)
1	base	5	0	46	0	n/a
1	log	6	0	46	0	none
1	enforce	5	0	46	0	(not measured)
2	base	5	0	46	0	n/a
2	log	6	0	46	0	none
2	enforce	4	0	46	0	(not measured)
3	base	4	0	46	0	n/a
3	log	5	0	46	0	none
3	enforce	4	0	46	0	(not measured)

For the batch programs, "ready after" is the whole run (start to exit).

Result: PASS

4. What log mode found: syscalls the profiles miss

No operation failed in any mode, so every application works under its enforced profile. Log mode shows, however, that five applications make syscalls outside their profile; under enforce these calls get EPERM and the application (or runc) carries on. The same syscalls appeared in all three repetitions.

Syscall	Applications	Made by	Why the profile misses it
fchdir	Tomcat, Jetty, Storm	the JVM (libjvm.so: PerfMemory::create_memory_region)	At start-up HotSpot creates its shared performance-data file (/tmp/hsperfdata_<user>) and changes into that directory with fchdir. This is JVM-internal code, not reached from the functions native methods are bound to; libjvm.so is analysed only from those (its no-binding run did not finish).
times	Solr	the JVM (libjvm.so: G1 GC-pause timing, TraceCPUTime)	Solr starts the JVM with GC logging, so G1 measures the CPU time of every pause with times(). This is JVM-internal code (the GC threads), not reached from the functions native methods are bound to (strace -k stack, control/strace_solr.txt).
fallocate	Storm	librocksdbjni (RocksDB, used by Storm's local state)	PosixWritableFile::Allocate is reached only through a virtual call. RocksDB's PosixEnv constructor is inlined and its vtable is loaded through a GOT entry, so SysPart never treats those functions as address-taken. RocksDB without binding reaches fallocate.

Syscall	Applications	Made by	Why the profile misses it
fchown	Elasticsearch, Jetty	runc (exe=/runc), while starting the container	Both images run as a non-root user; runc changes the owner of the container's stdio before it starts the program and ignores EPERM. runc is covered by the runc.txt whitelist, which has no fchown.

fallocate in Storm is a syscall that the no-binding analysis of the same library does reach (Part 2): here binding makes the profile too tight. fchdir and times are in libjvm.so, where no-binding is not resolved. fchown is outside the Java analysis (runc).

5. Problems found in the harness, and how they were fixed

• **Log-mode records were first read from the wrong place.** The first version read the kernel log (`journalctl -k`, type=1326). This host runs auditd, which takes the records, so they only appear in `/var/log/audit/audit.log` as type=SECCOMP. The positive control showed this. The harness now reads the audit log, and every log-mode run was redone (the first log runs are kept in results_journal_bug/). Base and enforce runs did not depend on this.

• **Jetty readiness.** The check requested /wl/, a JSP, but this Jetty image has no JSP support (HTTP 500 in every mode, including base). It now checks a static file, and the workload requests static files.

• **Storm readiness.** nimbus writes its log to a file, not to stdout, so the log-based check never fired (in every mode, including base). It now waits for the Thrift port 6627.

• **Smaller workload fixes** (each also failed under base): Solr query encoding and the backup location; a Groovy GString; a JRuby Range#sum bug inside threads (replaced by a loop); root-owned Tomcat and Jetty work directories (removed with sudo between runs).

• **Storm profile.** The first rerun used the 173-syscall profile file; the paper reports 172 (without sync), so Storm was rerun with that profile (the 173 runs are kept in results_storm_173/).

6. After the fixes: the same 90 runs with the fixed profiles

Three analysis fixes close the four gaps of section 4. (1) libjvm.so makes every syscall through functions it imports, so SysPart also runs on each library libjvm imports from (libc, libpthread, libdl, libm, ld-linux), starting from only those functions: this covers the JVM's own start-up code and threads (fchdir, times). (2) For C++ libraries other than libjvm, every function a vtable points to becomes a start function, so virtual calls through a vtable loaded via the GOT are followed (RocksDB: 59 to 70 syscalls, including fallocate). (3) fchown is added to the runc whitelist. Every other library result is unchanged. The positive control, rerun between these runs, again reported exactly mkdir and rename.

Application	Profile before	Profile after	Added syscalls	Failed ops (9 runs)	Outside the profile (log mode)	Result
Apache ZooKeeper	190	192	rt_sigsuspend, times	0	none	PASS
Apache Tomcat	170	173	fchdir, rt_sigsuspend, times	0	none	PASS
Apache Cassandra	230	230	none	0	none	PASS
Eclipse Jetty	157	161	fchdir, fchown, rt_sigsuspend, times	0	none	PASS
Apache Storm	172	179	fallocate, fchdir, fchown, readahead, rt_sigsuspend, sync_file_range, times	0	none	PASS
OrientDB	173	174	fchown	0	none	PASS
Elasticsearch	183	185	fchown, rt_sigsuspend	0	none	PASS
Apache Solr	199	201	rt_sigsuspend, times	0	none	PASS
Apache Groovy	189	189	none	0	none	PASS
JRuby	167	168	fchown	0	none	PASS

Mean profile size: 183.0 before, 185.2 after. Runs: results_fixes/.

Part 2. Binding vs no binding (static analysis, all ten images)

What is compared. For each container, SysJava analysed a set of native libraries (e.g. 14 in Tomcat 7). For each library SysPart builds a call graph and collects the syscalls reachable from its start functions. **With binding**, the start functions are only the C functions the application's native methods are bound to (JNI names, RegisterNatives tables, JNA/JNR/FFM symbols): this is the pipeline's own run. **Without binding**, the start functions are every function in the library, because without knowing the binding any of them could be called. The library file and the SysPart build are the same; only the start functions change.

Identical library files across images (the shared JDK libraries, libc) are run once: 141 library analyses in 10 images are 38 distinct files, so 37 no-binding runs (libjvm.so excluded: its no-binding run ran out of memory at 35 and 60 GB on Tomcat 11, so it is "not resolved" and keeps its binding set in the profile column).

Application (JDK 8u504, Debian 10)	Libraries	Binding: starts / syscalls	No binding: starts / syscalls	Profile with binding	Profile without binding	Added
Apache ZooKeeper 3.4.14	11	328 / 108	3,648 / 128	190	200	+10
Apache Tomcat 7.0.109	14	630 / 124	4,686 / 138	170	178	+8
Apache Cassandra 3.0.29	17	750 / 124	8,261 / 275	230	304	+74
Eclipse Jetty 9.4.51.v20230217	8	191 / 92	1,978 / 120	157	171	+14
Apache Storm 2.4.0	14	1,637 / 125	12,707 / 144	172	185	+13
OrientDB 3.2.19	17	487 / 111	4,747 / 129	173	183	+10
Elasticsearch 7.5.2	15	461 / 112	7,334 / 275	183	295	+112
Apache Solr 8.7.0	18	809 / 127	5,809 / 140	199	204	+5
Apache Groovy 4.0.0	15	375 / 109	3,990 / 128	189	198	+9
JRuby 9.4.2.0	7	430 / 95	1,382 / 119	167	177	+10
Mean				**183.0**	**209.5**	+26.5

"Starts / syscalls" are over all libraries except libjvm.so; the profile columns are the full profile (the published one, and the same with every library's no-binding syscalls added).

Why Cassandra and Elasticsearch grow most. Both call libc directly through JNA. With binding, SysPart starts from the 3 (Cassandra) or 5 (Elasticsearch) libc functions the Java code binds to (3 and 37 syscalls). Without binding, all 3,125 libc functions are starts and 273 syscalls are reachable, including ptrace, mount, init_module, reboot, swapon and pivot_root. In the other eight images the increase is 5 to 14 syscalls, mostly inotify_*, lchown, mknod, symlink, link and getitimer. In every library, no-binding reaches every syscall binding does.

Reproduce

Workload validation, from the artifact (analysis/workload_validation; needs Docker, sudo for the audit log; its README describes every workload, and results/ holds every run's JSON, container log and raw audit records):

```
python3 control/positive_control.py control/result_check.json   # must print "passed": true
export WLV_PROFILE_DIR=$PWD/profiles/fixed WLV_RESULTS=results_rerun   # or profiles/paper
python3 run_validation.py zookeeper tomcat cassandra jetty storm orientdb elasticsearch solr groovy jruby --reps 3
python3 make_table.py          # $WLV_RESULTS/table.md
```

Binding vs no binding, from the artifact (analysis/binding_ablation/jdk8_all):

```
python3 unpack_libs.py          # symbolized libraries of the 10 images (inputs/libs.tar.xz)
python3 make_targets.py         # 141 analyses -> 38 distinct files -> 37 no-binding runs
python3 compare_all.py          # tables from the shipped no-binding results
# full rerun of the 37 SysPart runs (hours, up to 24 GB per run):
SYSPART_APP=/path/to/SysPartCode/analysis/app JOBS=2 bash run_all.sh && python3 compare_all.py
```